

7.3 正变化方法与常用正交多项式

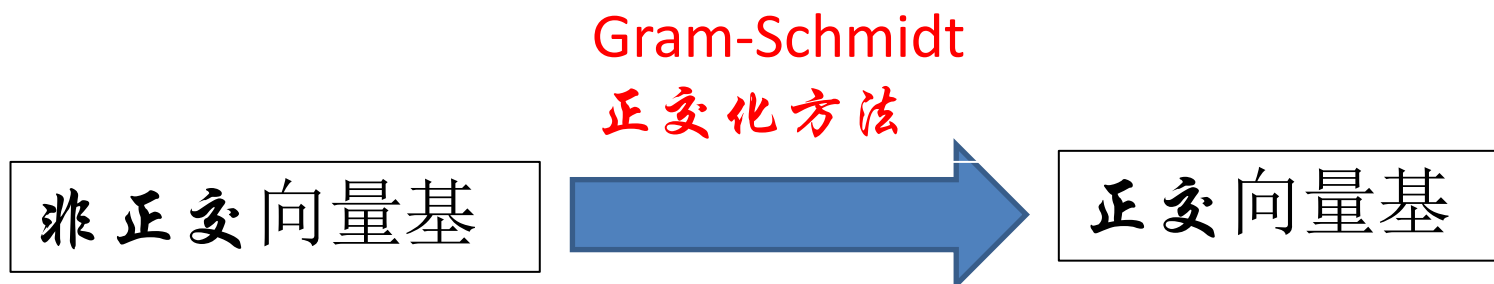
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7.3 正变化方法与常用正交多项式

- 7.3.1 Gram-Schmidt 正变化方法（补充内容）
- 7.3.2 常用正交多项式

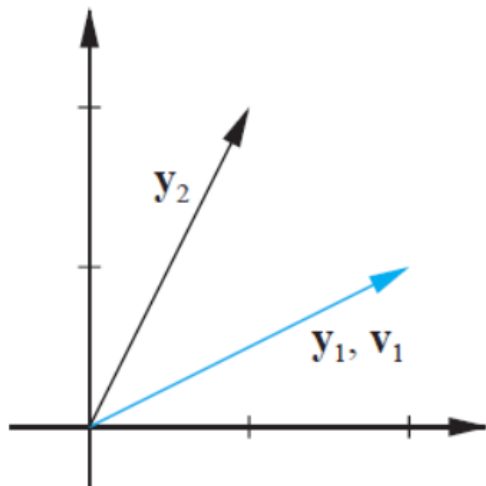
7.3.1 Gram-Schmidt 正变化方法



Gram (1850-1916) 丹麦数学家

Schmidt (1876-1959) 德国数学家

$$y_1, y_2, \dots, y_n \quad \longrightarrow \quad v_1, v_2, \dots, v_n$$

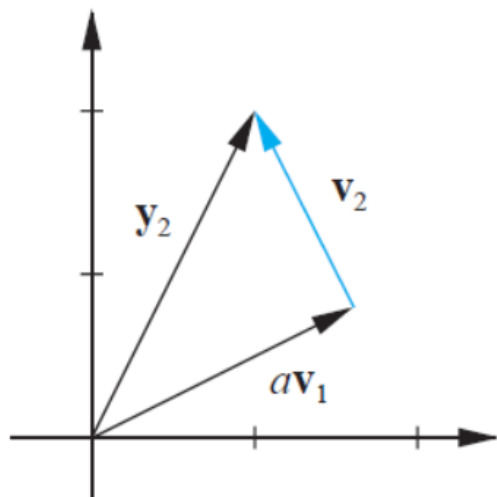


$$v_1 = y_1$$

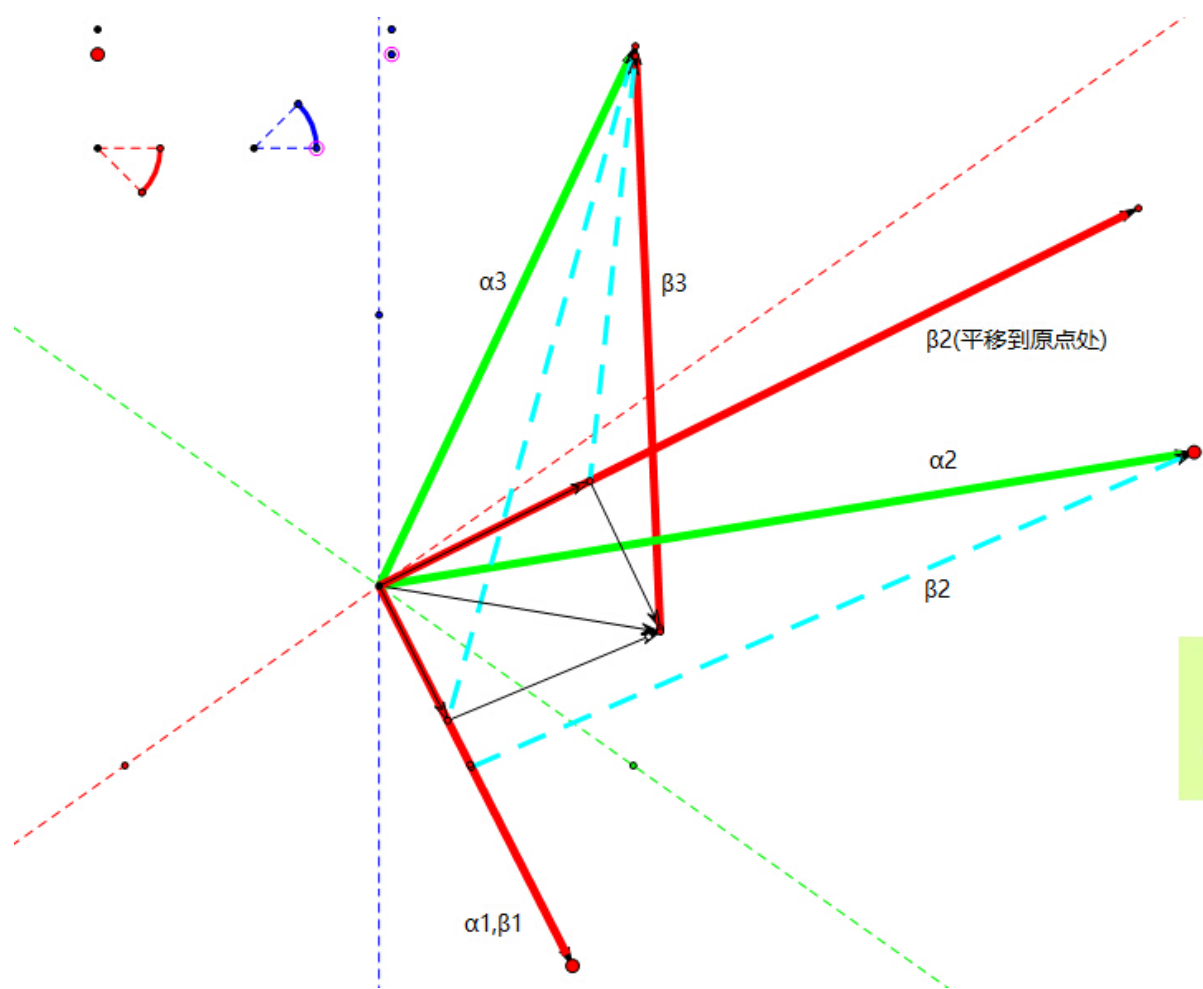
$$v_2 = y_2 - a v_1$$

$$(v_1, v_2) = (v_1, y_2 - a v_1) = (v_1, y_2) - a(v_1, v_1) = 0$$

$$a = \frac{(v_1, y_2)}{(v_1, v_1)}$$



$$v_k = y_k - \sum_{i=1}^{k-1} \frac{(v_i, y_k)}{(v_i, v_i)} v_i$$

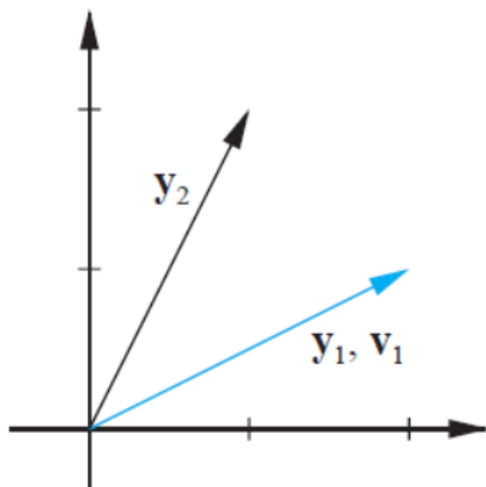


$$\beta_1 = \alpha_1$$

$$\beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1$$

$$\beta_3 = \alpha_3 - \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)} \beta_1 - \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)} \beta_2$$

举例说明

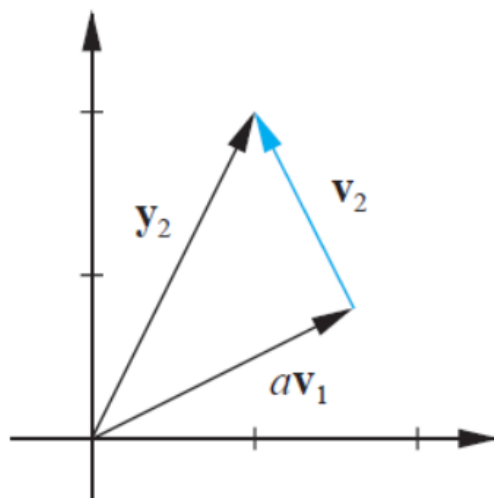


$$\mathbf{y}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{y}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

$$\mathbf{v}_1 = \mathbf{y}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\mathbf{v}_2 = \mathbf{y}_2 - \frac{\mathbf{v}_1^T \mathbf{y}_2}{\mathbf{v}_1^T \mathbf{v}_1} \mathbf{v}_1$$

$$= \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \frac{\begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}}{\begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 1.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} -0.6 \\ 1.2 \end{bmatrix}$$



7.3.2 常用正交多项式

只要给定**区间** $[a, b]$ 及**权函数** $\rho(x)$ 均可由一族线性无关的幂函数 $\{1, x, \dots, x^n, \dots\}$, 利用Gram-Schmidt正交化方法, 可以构造出正交多项式序列 $\{\varphi_n(x)\}_0^\infty$

$$(f(x), g(x)) = \int_a^b \rho(x) f(x) g(x) dx.$$

$$v_k = y_k - \sum_{i=1}^{k-1} \frac{(v_i, y_k)}{(v_i, v_i)} v_i.$$

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

得到的正交多项式序列有以下性质：

(1) $\varphi_n(x)$ 是具有最高次项系数为1的 n 次多项式.

(2) 任何 n 次多项式 $P_n(x)$ 均可表示为

$\varphi_0(x), \varphi_1(x), \dots, \varphi_n(x)$ 的线性组合.

(3) 当 $k \neq j$ 时, $(\varphi_j(x), \varphi_k(x)) = 0$,

且 $\varphi_k(x)$ 与任一次数小于 k 的多项式正交.

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

(4) 存在递推关系

$$\varphi_{n+1}(x) = (x - \alpha_n)\varphi_n(x) - \beta_n\varphi_{n-1}(x) \quad (n = 0, 1, \dots).$$

其中

$$\varphi_0(x) = 1, \quad \varphi_{-1}(x) = 0,$$

$$\alpha_n = (x\varphi_n(x), \varphi_n(x)) / (\varphi_n(x), \varphi_n(x)),$$

$$\beta_n = (\varphi_n(x), \varphi_n(x)) / (\varphi_{n-1}(x), \varphi_{n-1}(x)), \quad (n = 1, 2, \dots).$$

这里 $(x\varphi_n(x), \varphi_n(x)) = \int_a^b x\varphi_n^2(x)\rho(x)dx.$

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

(5) 设 $\{\varphi_n(x)\}_0^\infty$ 是在 $[a, b]$ 上带权 $\rho(x)$ 的正交多项式序列,
则 $\varphi_n(x) (n \geq 1)$ 的 n 个根都是在区间 (a, b) 内的单重实根。

$$\varphi_0(x) = 1, \quad (f(x), g(x)) = \int_a^b \rho(x) f(x) g(x) dx.$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

7.3.2.1 勒让德多项式

当区间为 $[-1, 1]$, 权函数 $\rho(x) \equiv 1$ 时,

由 $\{1, x, \dots, x^n, \dots\}$

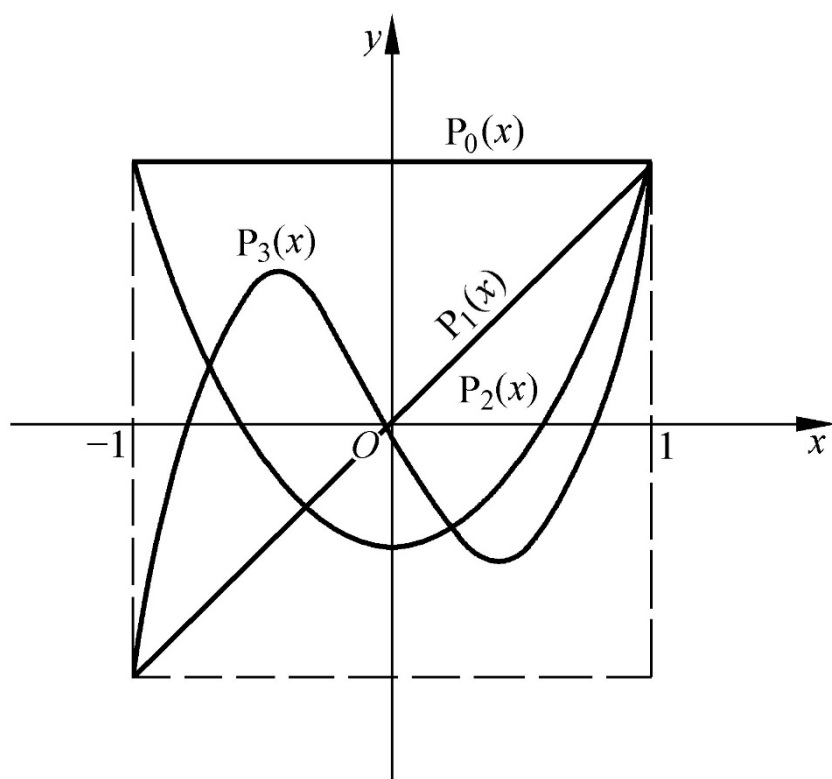
正交化得到的多项式就称为**勒让德**(Legendre)**多项式**,

并用 $P_0(x), P_1(x), \dots, P_n(x), \dots$ 表示.

罗德利克(Rodrigul) 给出了简单的表达式

$$P_0(x) = 1, \quad P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \{(x^2 - 1)^n\} \quad (n = 1, 2, \dots),$$

$$P_0(x) = 1, \quad P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \{(x^2 - 1)^n\} \quad (n = 1, 2, \dots).$$



$$P_0(x) = 1, P_1(x) = x,$$

$$P_2(x) = (3x^2 - 1) / 2,$$

$$P_3(x) = (5x^3 - 3x) / 2,$$

$$P_0(x) = 1, \quad P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \{(x^2 - 1)^n\} \quad (n = 1, 2, \dots).$$

勒让德多项式重要性质:

性质1 正交性

$$\int_{-1}^1 P_n(x) P_m(x) dx = \begin{cases} 0, & m \neq n; \\ \frac{2}{2n+1}, & m = n. \end{cases}$$

回顾

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

(4) 存在递推关系

$$\varphi_{n+1}(x) = (x - \alpha_n)\varphi_n(x) - \beta_n\varphi_{n-1}(x) \quad (n = 0, 1, \dots).$$

其中

$$\varphi_0(x) = 1, \quad \varphi_{-1}(x) = 0,$$

$$\alpha_n = (x\varphi_n(x), \varphi_n(x)) / (\varphi_n(x), \varphi_n(x)),$$

$$\beta_n = (\varphi_n(x), \varphi_n(x)) / (\varphi_{n-1}(x), \varphi_{n-1}(x)), \quad (n = 1, 2, \dots).$$

$$\text{这里 } (x\varphi_n(x), \varphi_n(x)) = \int_a^b x\varphi_n^2(x)\rho(x)dx.$$

性质2 递推关系

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$$

由 $P_0(x) = 1, P_1(x) = x$, 利用上述递推公式就可推出

$$P_2(x) = (3x^2 - 1)/2,$$

$$P_3(x) = (5x^3 - 3x)/2,$$

$$P_4(x) = (35x^4 - 30x^2 + 3)/8,$$

$$P_5(x) = (63x^5 - 70x^3 + 15x)/8,$$

$$P_6(x) = (231x^6 - 315x^4 + 105x^2 - 5)/16,$$

...

性质3 奇偶性

$$P_n(-x) = (-1)^n P_n(x).$$

性质4 $P_n(x)$ 在区间 $[-1, 1]$ 内有 n 个不同的实零点.

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

$$(f(x), g(x)) = \int_a^b \rho(x) f(x) g(x) dx.$$

7.3.2.2 切比雪夫多项式

当权函数 $\rho(x) = \frac{1}{\sqrt{1-x^2}}$, 区间为 $[-1, 1]$ 时,

由 $\{1, x, \dots, x^n, \dots\}$

正交化得到的正交多项式就是**切比雪夫(Chebyshev)多项式**.

它可表示为

$$T_n(x) = \cos(n \arccos x), \quad |x| \leq 1.$$

若令 $x = \cos \theta$, 则 $T_n(x) = \cos n\theta, 0 \leq \theta \leq \pi$.

$$T_n(x) = \cos(n \arccos x), \quad |x| \leq 1.$$

切比雪夫多项式有很多重要性质:

性质1 切比雪夫多项式 $\{T_k(x)\}$ 在区间 $[-1, 1]$ 上带权

$\rho(x) = 1 / \sqrt{1 - x^2}$ 正交, 且

$$\int_{-1}^1 \frac{T_n(x)T_m(x)dx}{\sqrt{1-x^2}} = \begin{cases} 0, & n \neq m; \\ \frac{\pi}{2}, & n = m \neq 0; \\ \pi, & n = m = 0. \end{cases}$$

$$T_n(x) = \cos(n \arccos x), \quad |x| \leq 1.$$

性质2 递推关系

$$T_0(x) = 1, \quad T_1(x) = x.$$

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \quad (n = 1, 2, \cdots),$$

由递推公式可推出：

$$T_2(x) = 2x^2 - 1,$$

$$T_3(x) = 4x^3 - 3x,$$

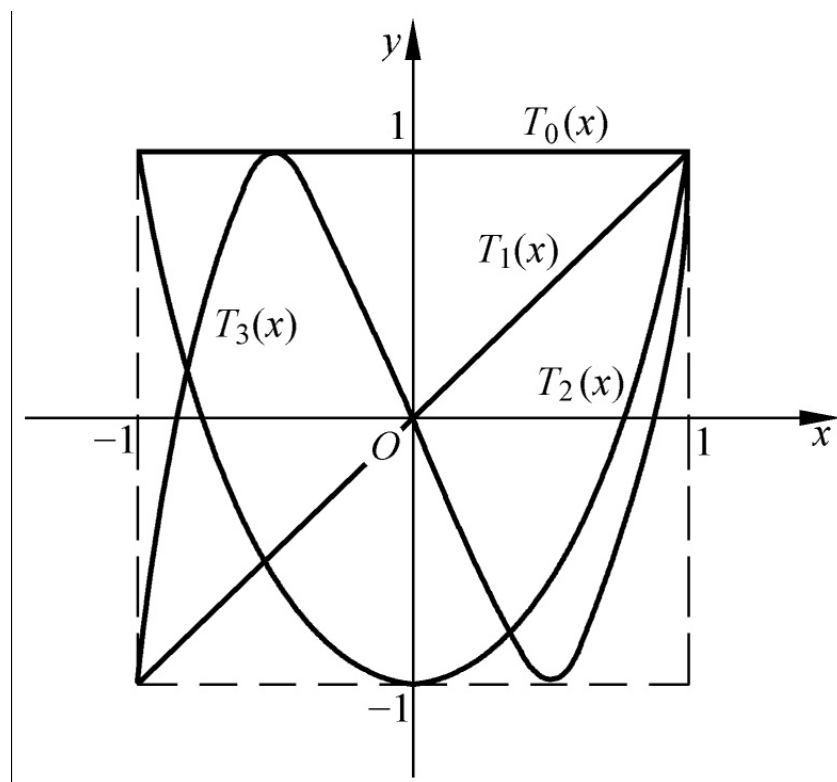
$$T_4(x) = 8x^4 - 8x^2 + 1,$$

$$T_5(x) = 16x^5 - 20x^3 + 5x,$$

$$T_6(x) = 32x^6 - 48x^4 + 18x^2 - 1,$$

...

$$T_0(x), T_1(x), T_2(x), T_3(x)$$



③ 奇偶性

$$T_n(-x) = (-1)^n T_n(x)$$

④ $T_n(x)$ 在 $(-1,1)$ 内的 n 个零点为 $x_k = \cos \frac{2k-1}{2n} \pi \quad (k=1,2,\cdots,n)$

在 $[-1,1]$ 内有 $n+1$ 个极值点, $y_k = \cos \frac{k}{n} \pi \quad (k=0,1,2,\cdots,n)$

⑤ $T_n(x)$ 的最高次幂 x^n 的系数为 $2^{n-1} (n \geq 1)$ 。

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

$$(f(x), g(x)) = \int_a^b \rho(x) f(x) g(x) dx.$$

7.3.2.3 其他常用的正交多项式

除上述两种最重要的正交多项式外，下面是三种较常用的正交多项式.

1. 第二类切比雪夫多项式

在区间 $[-1, 1]$ 上带权 $\rho(x) = \sqrt{1-x^2}$ 的正交多项式称为**第二类切比雪夫多项式**.

$$U_n(x) = \frac{\sin[(n+1)\arccos x]}{\sqrt{1-x^2}}.$$

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

$$(f(x), g(x)) = \int_a^b \rho(x) f(x) g(x) dx.$$

2. 拉盖尔多项式

在区间 $[0, +\infty)$ 上带权 $\rho(x) = e^{-x}$ 的正交多项式称为
拉盖尔(Laguerre)多项式.

$$L_n(x) = e^x \frac{d^n}{dx^n} (x^n e^{-x}).$$

$$\varphi_0(x) = 1,$$

$$\varphi_n(x) = x^n - \sum_{j=0}^{n-1} \frac{(x^n, \varphi_j(x))}{(\varphi_j(x), \varphi_j(x))} \varphi_j(x) \quad (n = 1, 2, \dots).$$

$$(f(x), g(x)) = \int_a^b \rho(x) f(x) g(x) dx.$$

3. 埃尔米特多项式

在区间 $(-\infty, +\infty)$ 上带权 $\rho(x) = e^{-x^2}$ 的正交多项式称为
埃尔米特多项式.

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}),$$

- End